

Dissertation

Tyler Wiederich

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1 CHAPTER 0

2 Chapter 0

Literature Review

3 Motivation and Background

Data visualizations play an essential role in understanding patterns in the structure of data. They allow for a programmatic mapping of data into a “picture” that can be used to gather insight from the viewer [tukey1965; tufte2001; wilkinson2005]. Since the early 20th century, researchers have been exploring the question: what defines a good graph? [croxton1927; croxton1932; cleveland1984; vanderplas2020]. Many attempts to answer this questions have provided good recommendations, but are largely limited to the projection of graphs onto 2-dimensional (2D) surfaces. While this addresses many of the typical use cases for data visualizations, graphs created in our 3-dimensional (3D) world are largely assessed.

Many charts exist within the confines of a 2D space due to the practicality of computer renderings. These computer-generated graphics are quick and cheap to produce [tukey1965], which possibly contributes to their widespread usage. Before computers, hand-drawn and complex printing techniques were common production methods and took time to produce [friendly2021; vanderplas2019]. In contrast, all data physicalizations require additional material resources, increasing the base cost to produce. This results in the majority of modern graphics being produced with 2D representations on digital screens.

Nearly every type of chart can be created by using a programmatic mapping of data and variables to aesthetics and geometries, a process @wilkinson2005 referred to as the “Grammar of Graphics.” This has found widespread implementation into numerous software [@wickham2010; @satyanarayan2017; @sas9.4]. For example, consider the chart created in Figure Figure 3.1. In this figure, the Palmer Penguins dataset [@penguins] is supplied to the `ggplot` function from the `ggplot2` package [@ggplot2]. The x and y axes are mapped to bill length and bill depth, with color and shape being mapped to island. The geometry is given by `geom_point`, which results in a scatter plot that shows each complete data observation. Although the chart would benefit from further customization, it was quick to produce and has instant visual insights.

```
library(tidyverse)

library(palmerpenguins)

ggplot(penguins, mapping = aes(x = bill_length_mm,
                               y = bill_depth_mm,
                               color = island,
                               shape = island)) +

  geom_point()
```

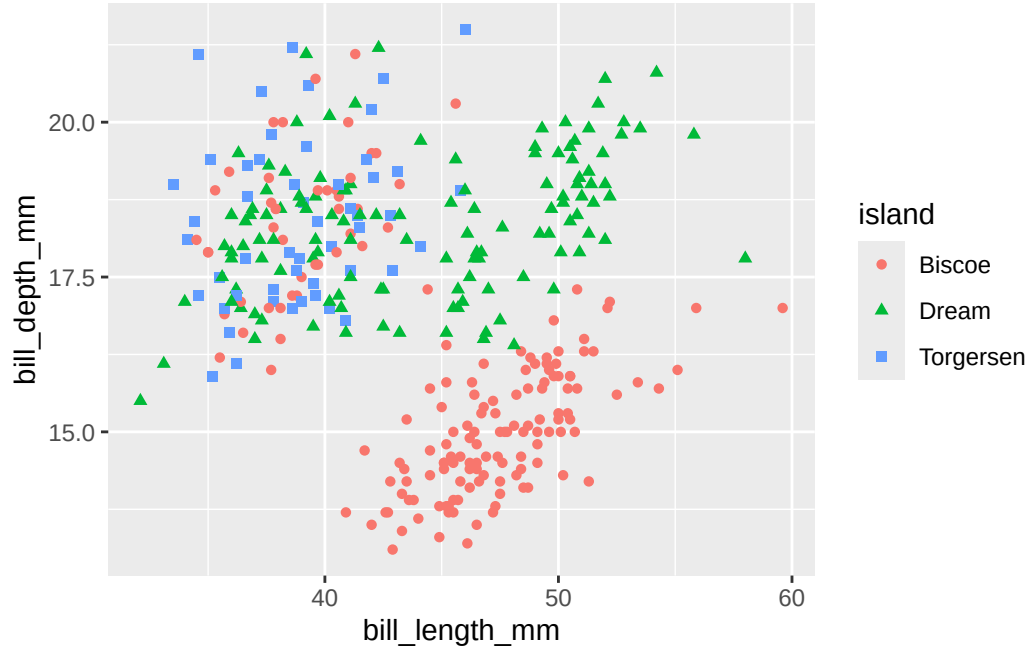


Figure 3.1: Example of a graphic created using the grammar of graphics.

While there have been many attempts to establish good data visualization practices [tufte2001; vanderplas2020], we do not know how well these recommendations translate from 2D projections into our 3D world. The advent of 3D printing allows us to precisely construct the 3D charts that have widely been studied through their 2D representations [croxton1932; barfield1989; zacks1998]. With limited research on 3D-printed data manifestations, we will explore the initial empirical findings of true 3D statistical graphics.

4 Overview of Physical 3D Visualizations

4.1 Creating 3D-Printed Visualizations

The popularity of the 3D printer exploded in the early 2010's as the technology became increasingly available and cheaper. This has led to increase usage of 3D printing in many focus areas, including healthcare [dodziuk2016] and engineering [v2023], but has only seen novel use cases for statistical graphics. Many software programs have the ability to create 3D statistical graphics, but lack the ability to easily export the graphics into files suitable for 3D printing.

Numerous software programs have the can create renderings of 3D charts. Excel [msexcel] has native support for creating charts with 3D depth cues, but requires add-ins to produce charts using 3 axes. R (XXX) has a several options for creating 3D charts that have data on 3 axes [e.g., @rgl; @rayshader; @plotly], but lacks support for creating depth charts. SAS 9.4 (XXX) has options such as PROC GCHART for adding depth cues and PROC G3D for surface charts. Other popular software programs contain similar capabilities, such as JavaScript libraries, MATLAB, Python, and more.

While a number of tools exist for creating 3D data visualizations, the pipeline of getting

these charts into files compatible with 3D printing is not widely automated. For example, R has some options using the `rgl` [rgl] and `rayshader` [rayshader] packages, but both packages have limited tuning parameters for the resulting output files. Another option is to use 3D software, such as OpenSCAD [kintelOpenSCADDocumentation2023], to manually create the charts, but these software programs lack the ability to directly integrate statistical information.

Another consideration for 3D-printed charts is the inclusion of text labels. With single-filament 3D printers, text can be incorporated in three primary ways: (1) embossing, in which the text is raised above the surface of the print; (2) engraving, in which the text is recessed into the surface; or (3) applying external labels, such as stickers or ink. Multi-filament 3D printers have the additional option to use a different color to make text labels flush with their surfaces. However, @munzner cautioned that the text of 3D charts will suffer from legibility issues due to the distortion of the text when viewed at an angle.

FIGURE PLACEHOLDER: examples of implementations of text on the charts

4.2 Current use cases

The use of 3D-printed data visualizations for statistical graphics is largely a novelty in the early 21st century. Services such as WhiteClouds (www.whiteclouds.com) offer the ability to create a wide variety of 3D-printed graphics, and more “do-it-yourself” options can be achieved with \$300-\$500 in start-up costs. After acquiring a 3D printer, 1 kilogram rolls of PLA filament cost approximately \$10 on Amazon, resulting in relatively inexpensive charts.

FIGURE PLACEHOLDER: show some examples of 3D printed charts

5 Bar Charts in Research and Practice

One of the most common types of charts is the bar chart, characterized by the use of rectangular geometries. Two variables are mapped to the geometries of the bar chart, where one axis is dedicated to a categorical variable of nominal or ordinal nature, and the other axis is for a continuous variable to denote magnitude or a response. These charts have been used over 200 years, credited to William Playfair in his visualization of trade data [friendly2021].

The use of 3-dimensional elements in bar charts typically involve converting the rectangular bars into rectangular prisms. The naming convention for these types of charts is not consistent, sometimes referred to as 2.5D charts when the additional dimension does not convey extra information [e.g., tractinsky1999] or, more commonly, as 3D bar charts [e.g., zacks1998; fischer2000]. All following references to this style of chart will be called “3D bar charts.”

5.1 Against 3D Bar Charts

tuft2001 called the use of 3D elements a “fake perspective” in his description of “chart junk”, which is where visual elements add clutter the data visualization. Other researchers

[@zacks1998; @stewart2009] have termed the perspective effect “extraneous” when referring to depth cues. The reasoning behind this is fairly intuitive: why include additional noise when simpler alternatives exist?

The seminal work of @cleveland1984 theorized that encoding information in volumes would provide worse numerical estimation than for encodings using positions, lengths, and areas. Their reasoning stems from Steven’s Power Law (XXX), a psychophysics formulation that estimates a perceived stimuli magnitude p with the actual magnitude a by $p = ka^\alpha$. The estimation of α provides guidance to what types of stimuli are subject to the most distorted to their true scale when $\alpha = 1$. (alpha for volume estimated to be X, area is Y, and length is Z). While the 3D bar charts do not lose their 2D encodings, they do gain depth, and thus volume encodings. The effect of volume versus area comparisons was noted earlier by @croxton1932, where metrics of accuracy were the worst for cubes as compared to bars and circles.

5.2 For 3D Bar Charts

@levy1996 made an argument for the use of depth cues in certain situations. In a series of three experiments, @levy1996 tasked over 100 psychology students with selecting one of various 2D and 3D charts that they thought would be best for certain scenarios. What they found was that students tended to pick 3D charts over 2D charts when needing to increase memory recollection of the chart, and about equal preference for 2D and 3D charts when presenting the information to others. @levy1996 theorized that the depth embellishments in the graphic may increase memorability of the chart by making the chart stand out.

Other types of “chart junk” have shown that additional non-data features can increase

recollection of bar charts at the cost of needing longer processing times [bateman2010; borgo2012; peña2020]. This increase in time has been shown for 3D bar charts [siegrist1996; fischer2000; stewart2009], lending some credibility to the argument that 3D depth cues could increase memorability.

5.3 Conflicting Empirical Studies

When accuracy is the goal, 2D is not better than 3D in some cases

Although tuftes2001 argued against the use of 3D charts when possible, empirical studies involving direct comparisons of 2D and 3D bar charts have shown mixed results. In general, these studies focus on metrics of accuracy and find that 3D bar charts tend to be either less accurate or just as accurate as their 2D counterparts. Another common metric is response time, where 3D charts tend to have longer response times. Overall preference for one style of chart over another is a subjective measurement, but has shown mixed results for the two styles of charts.

5.3.1 Accuracy

Accuracy is a common metric in studies comparing 2D and 3D bar charts, typically measured by either extracting a single numeric estimate or making comparisons between two values. Many studies used controlled environments for constructing stimuli, often displaying one or two stimuli at a time, or controlling non-target stimuli. In practice, charts are often more complex, displaying many stimuli without drawing attention to particular values.

When only measuring accuracy, 3D bar charts tend to be less accurate than their 2D

counterparts [zacks1998; stewart2009]. However, this result is less prominent when introducing time delays [zacks1998, Experiment 2] or easier task complexity [stewart2009]. siegrist1996 noted that bar positions and sizes also affected performance accuracy, emphasizing that other factors contribute to accuracy. It is also important to note that metrics of accuracy are not conclusive, where sometimes there is no evidence of a difference between 2D and 3D bar charts [melodycarswell1991].

Of course, there are other less-studied factors that could affect perceptual judgments of accuracy in 3D bar charts. In addition to viewing parameters found in 2D bar chart studies [fischer2005; rice2024], adding depth cues are also subject to viewing angles, amount of depth, and field of view parameters. Viewing angle, with respect to the axis, can mislead the reader into overestimating or underestimating numerical quantities (FIGURE X). zacks1998 did not find a significant effect for the amount of perception depth, but noted a trend such that increased exaggeration of depth lowered their accuracy metric. Lastly, field of view is fixed for the viewer, but can widely distort 3D renderings (FIGURE X). These factors add additional complexity, but are not widely studied for the use of statistical graphics.

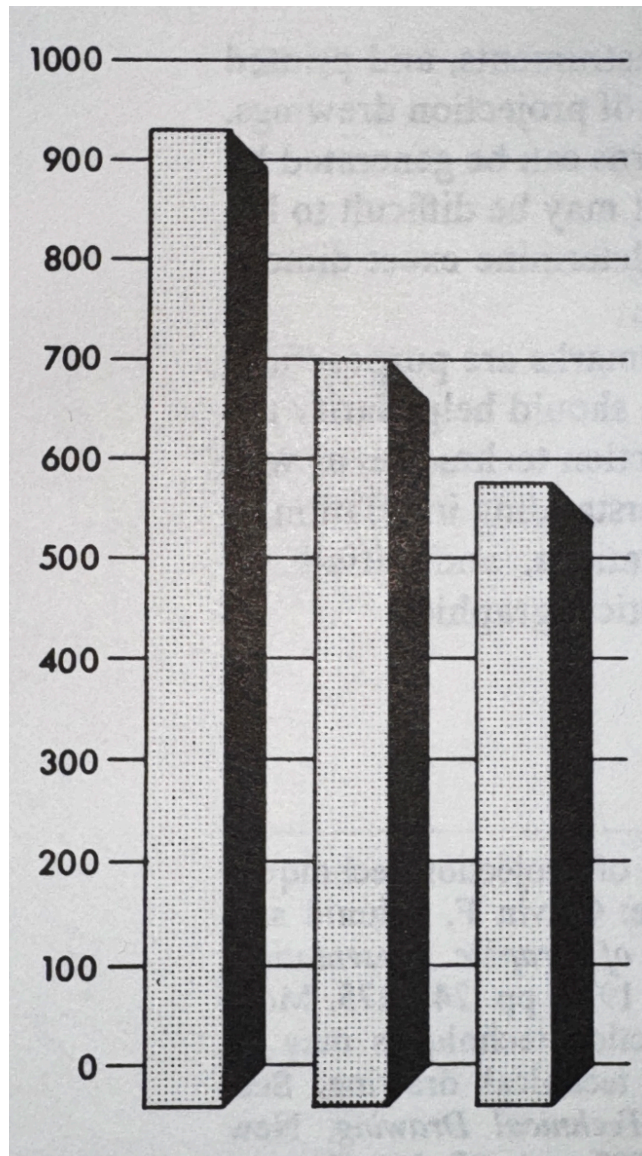
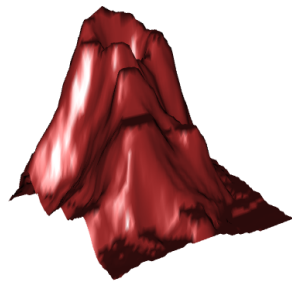


Figure 5.1: Figure in Statistical Graphics: Design Principles and Practices. The viewing angle for this chart would have viewers overestimate the values of the bars

5.3.2 Response Times

The speed in making perceptual judgments is another common metric in studies comparing types of charts. In theory, charts where questions can be answered more quickly implies that the chart is better at communicating information. However, response time is a poor



(a) FOV: 30 degrees



(b) FOV: 60 degrees



(c) FOV: 90 degrees



(d) FOV: 120 degrees



(e) FOV: 150 degrees

Figure 5.2: Multiple charts of the volcano dataset [`@volcano`] rendered using the `rgl` package [`@rgl`] in R. Each chart has a different field-of-view parameter, with the amount of distortion increasing as the field-of-view (FOV) increases.

metric in situations of exploratory data analysis and long term interactions with complex graphics [vanderplas2020]. When a question of interest is known in advance, response time becomes a more viable metric.

Depth cues from 3D bar charts tend to increase the amount of time required to answer questions [siegrist1996; fischer2000; stewart2009]. Intuitively, additional complexity should require additional time to process. For 3D bar charts, viewers need to align the response axis with the rectangular prisms in order to determine the approximate magnitude of the bar. Consider Figure X, where the y-axis denotes magnitude with respect to a hidden face of the bar. In order to extract a numerical quantity, the viewer first has to align the axis with the correct face of the prism before processing the magnitude.

5.3.3 Preference

The debate between whether to use 2D or 3D graphics stems from a wider adoption of 3D charts in practice. Computer graphics have made it easy to produce charts of either dimensionality, resulting in creative liberties for the creation of the chart for publication. Figures X and X showcase two examples of 3D bar charts from the 1978 Handbook of Agricultural Charts (XXX) where other options exist for displaying the data. [schmid1983 (Ch. 8)] also showcases many other 3D charts published in official reports.

U.S. SUGARBEET AND SUGARCANE PRODUCTION

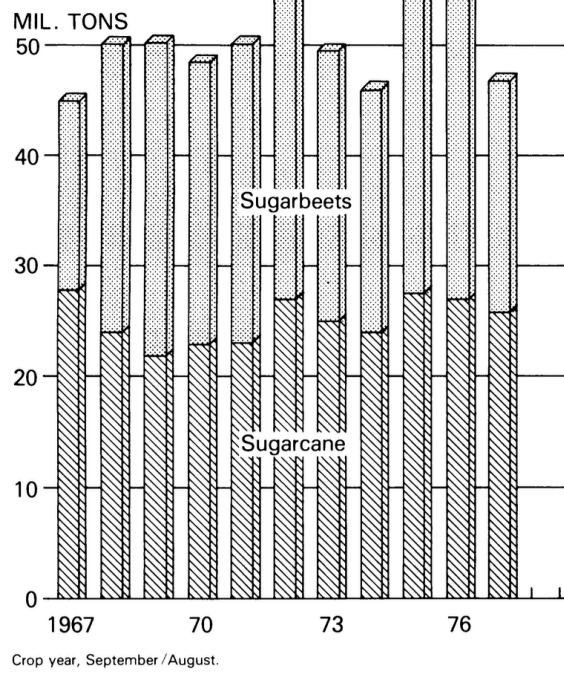
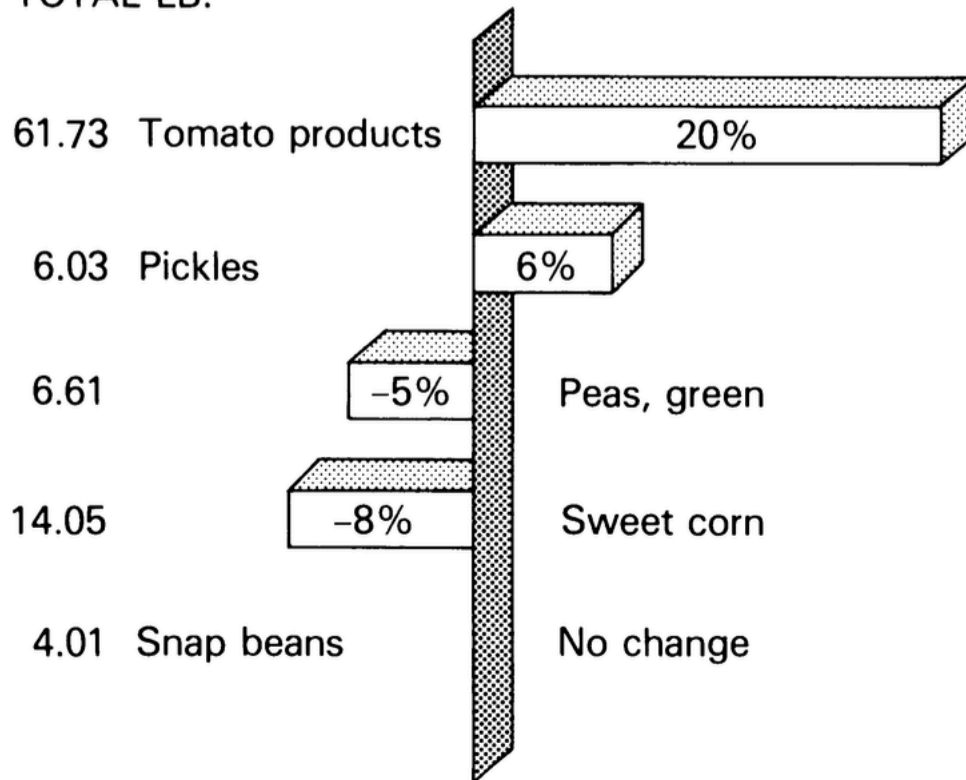


Figure 5.3: Located from 1978 HANDBOOK OF AGRICULTURAL CHARTS

CHANGES IN CANNED VEGETABLE CONSUMPTION PER CAPITA BETWEEN 1970-72 AND 1975-77

TOTAL LB.

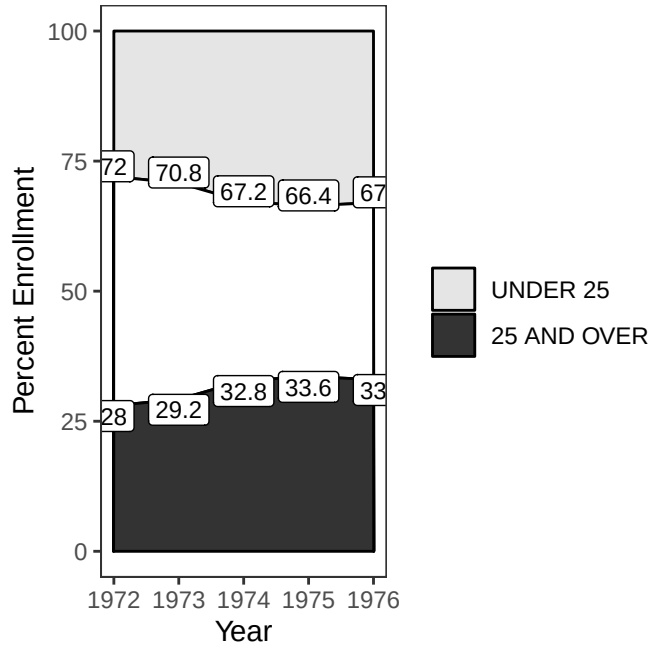


Fresh-weight basis.

Preference can be broken down into two broad categories: (1) visual appeal, and (2) usefulness. Visual appeal is where perceptual tasks are not a priority, and the goal is to have an image that is pleasant to the viewer. Usefulness, on the other hand, is a rating of how effective the viewer thought the chart was in order to answer the question. Both are subjective measures

When not tasked with estimating values, 3D bar charts tend to be more visually appealing [Levy1996; Fisher1997; Stewart2009]. This may be in part due to the forced illusion of depth which creates additional embellishments that would increase the impression and

memorability of the chart, as seen by other types of Tufte’s “chart junk” [borgo2012; peña2020]. Despite the visual appeal, 2D charts tend to be chosen for preference and ease of use when making numerical estimations [levy1996; stewart2009].



5.4 Construction of 3D Bar Charts

All of the aforementioned studies have one key limitation: they are conducted in environments where the 3D bar charts are projected onto 2D surfaces. An argument could be made that tactile charts created for persons with vision impairments could be considered 3D [goncu2008; engel2017; watanabe2018], but current implementations are vastly different from the typical design of 3D bar charts.

6 Heat Maps in Research and Practice

When creating charts for three or more variables, it becomes increasingly complex to visualize data [grinstein2001]. One option is to map the additional variables using gestalt principles [vanderplas2020] (Original citation?), which includes making use of color, size, and/or shapes to denote information. However, not every aesthetic is compatible with every type of variable. Color can be used to map either continuous variables with gradient scales or categorical variables with palettes. Size is typically reserved for continuous variables, where larger values correspond to a larger quantity. Shapes tend to have finite distinguishable representations in order to represent categorical variables. Examples of good and poor cases for each of these aesthetics is presented in Figure Figure 6.1.

Heat maps are a type of chart that can be used with two variables of any type and a continuous response variable. In two-dimensional formats, color is often used to denote the magnitude of the response, utilizing a gradient scale. When represented in three dimensions, height is the primary mapping of the response of the heat map. Due to the height aesthetic, 3D heat maps are sometimes referred to as “height maps”, particularly when spatial coordinates are involved.

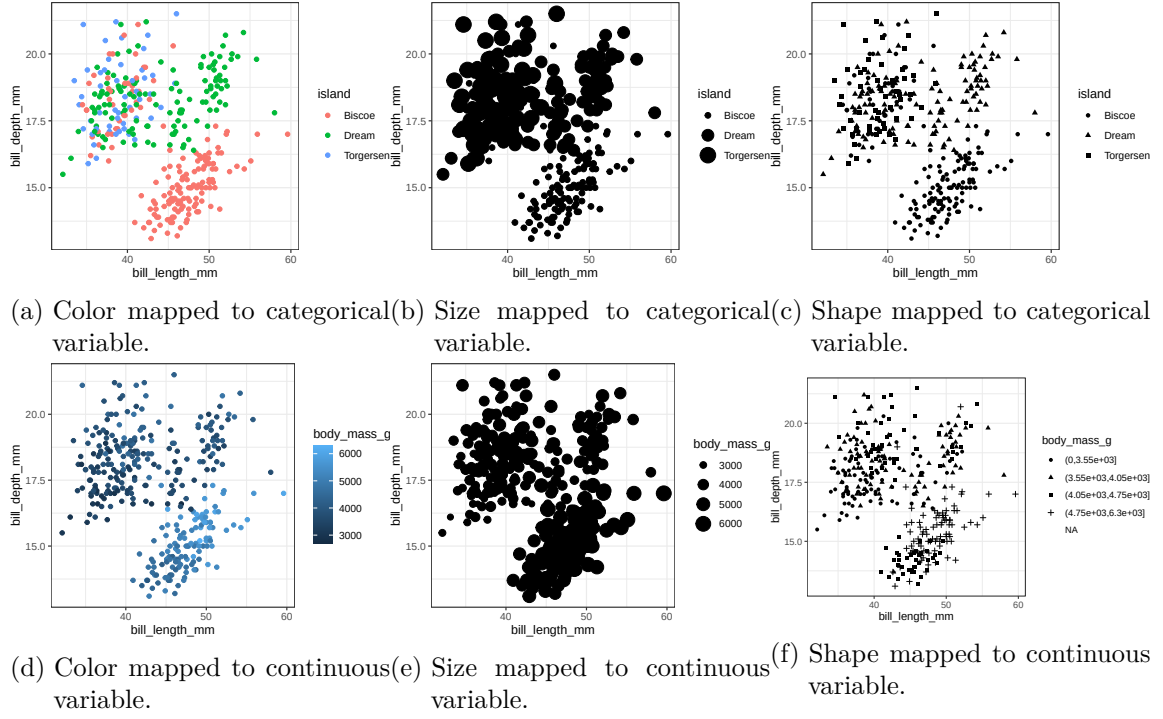


Figure 6.1: Examples of color, size, and shape for scatterplots of the penguins dataset [penguins].

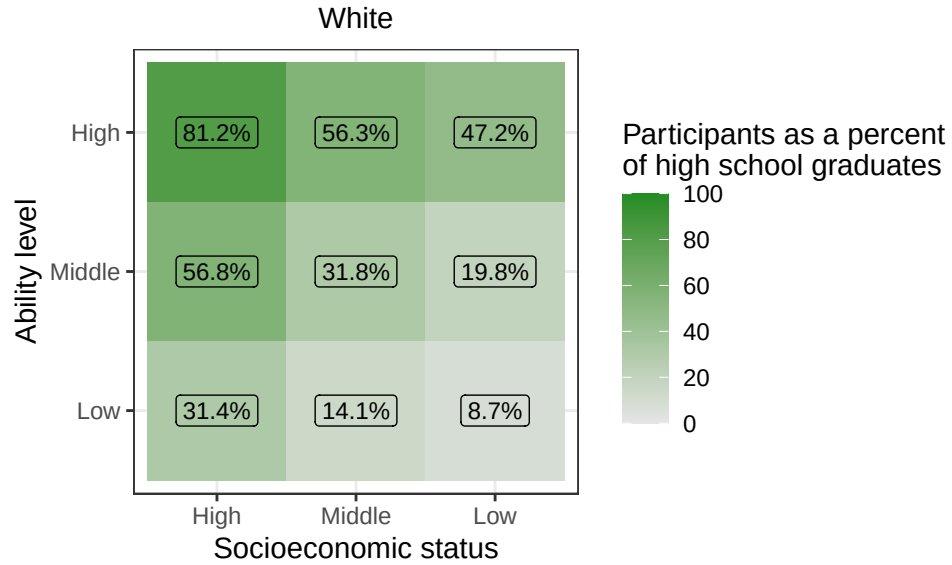
6.1 Empirical Studies of 2D and 3D Heat Maps

Unlike 2D and 3D bar charts, the comparison of dimensionality for heat maps has not been widely studied. This may in part be due to the complex nature of color scales [breslow2009; liu2018; reda2019]. While bar charts simply add a perspective depth cue to increase the dimensionality, the translation of color to height results in numerous considerations, such as the scaling of the height (Figure fig-color-height).

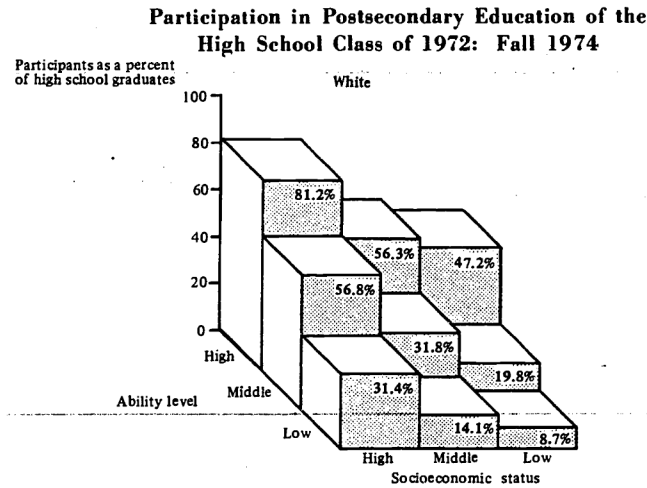
The type of media can affect the perception of 2D and 3D heat maps [kraus2020; jeong2025].

- @barfield1989
- @kraus2020

Participation in Postsecondary Education of the High School Class of 1972: Fall 1974

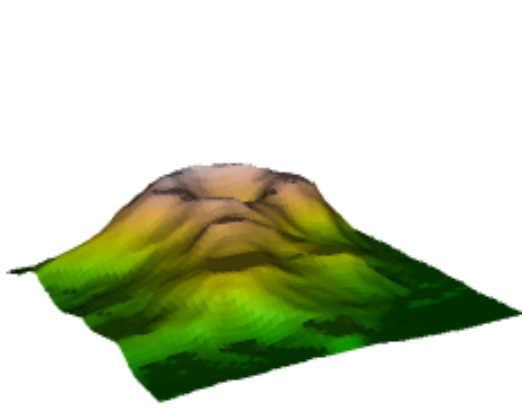


(a) 2D heat map



(b) 3D heat map

Figure 6.2: Heat maps created from with data from (XXX). The 2D heat map represents the percentage of high school graduates using a gradient color scale, whereas the 3D heat map uses height.



(a) Flatter height scale



(b) Stretched height scale

Figure 6.3: Two figures representing the volcano dataset [volcano] using different height scaling factors. Both figures use the same color scale, but there is no “correct” translation of color into the measurable height.

- @jeong2025 would be good, but uses VR
- @casali1988

7 Methods for Evaluating Visualization

Effectiveness

7.1 What makes a good chart?

@vanderplas2020 discussed various approaches to testing graphics, including numerical estimations, speed, and error rates. These metrics are fairly common metrics when comparing 2D and 3D charts [@croxton1932; @cleveland1984; @hughes], but do not represent the full picture of what makes a one type of chart better than another. As seen by @levy1996, the objectives of the chart can determine which type of chart should be used.

Since true 3D charts have limited research compared to 2D representations, it is reasonable to compare how perceptual judgements of numerical estimations compare to earlier studies in statistical graphics.

7.2 Designing studies for numerical estimations

Over 2,500 citations have referenced the work of @cleveland1984 when discussing properties of statistical graphics. Since then, the methods of analyzing and conducting studies

on statistical graphics have evolved [e.g., @vanderplas2020; @hofmann2012], but yet the structure of experimental designs involving numerical estimations tend to remain similar. Tasks typically involve the comparison of two stimuli, where the question is “how much bigger/smaller is one value compared to the other?” Care needs to be taken in the phrasing of this question since mental subtractive processes are perceived different from mental ratio processes [@veit; @hagerty1978].

@cleveland1984 asked participants two questions when making comparisons between two values on a chart: (1) which of the two values was the smaller, and (2) what percentage the smaller was of the larger. The first question clarified if the participants were looking at the values correctly for the second questions, and the second question is an estimation of A/B , where $A < B$. Accuracy was measured by $\log_2(|\text{True value} - \text{Estimated value}| + 1/8)$, using the trimmed averages when bootstrapping confidence intervals. Cleveland and McGill’s experiment was replicated by @heer2010, showing similar results for the experiment’s procedure in a crowdsourced environment.

When designing studies for numerical estimation, the choice of A/B and the measure of error are important considerations. Many researchers seemingly choose ratios in an arbitrary fashion or follow previously established work [e.g., @zacks1998 used the same ratios as @melodycarswell1991]. As for measures of error, several formulations can be used.

- $|\text{Error}|$ for magnitude of error
- Error for direction of error
- $\log |\text{Error}|$

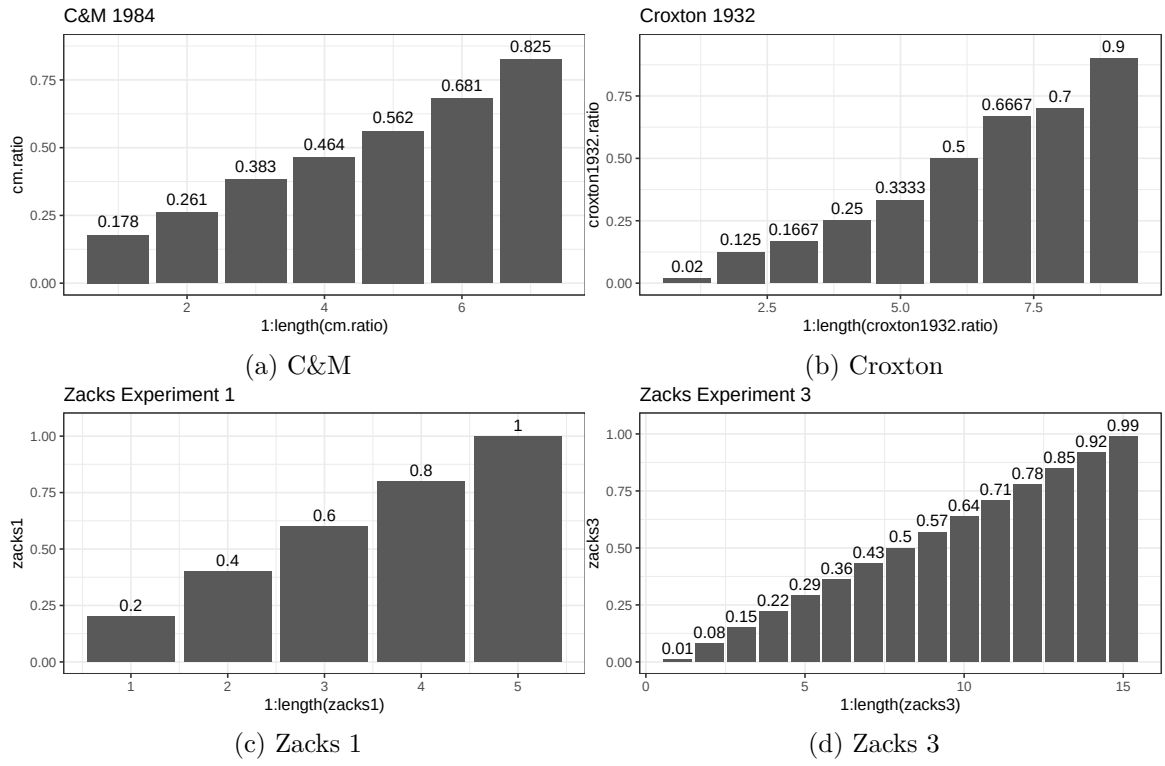


Figure 7.1: Ratios used by several studies

7.2.1 References

8 CHAPTER 2

9 Title: Using a 3D graphics experiment for experiential learning in an introductory statistics course.

10 Abstract

A key component of statistics courses is to teach students how to interpret data visualizations. Although many research-based recommendations exist for creating graphs, the technological advances for creating such graphs have outpaced studies that evaluate their effectiveness, especially when considering 3D graphs. Here, we describe a process for integrating an experiment on 3D graphs as part of a project on statistical investigations for students enrolled in an introductory statistics course and gathered responses as students reflected on their experience of being a participant in the experiment and then being a reviewer of empirical evidence about the experiment. A total of 82 students participated in our graphics project. As participants in an experiment, they displayed a pattern of not fully grasping research objectives as evidenced by widely varied responses. However, as reviewers of material from a pilot study of the same design, they tended to gain a clearer understanding of the purpose of the experiment and its role in the realm of data visualizations by correctly interpreting an extended abstract and video presentation of the pilot study. The project we presented to students shows promise as an educational tool for helping students gain a more holistic view of statistical research, which is important in both the contributions to data visualizations and the education of statistics.

11 Introduction

The education of statistics emphasizes the use of real data and its application in answering research questions. Students in introductory statistics courses are mainly exposed to elementary methods and textbook examples that demonstrate the application of these methods, which is used, in part, because of the emphasis on teaching students to think statistically using real data [carver]. Many textbooks, such as Tintle2021, use scenarios and ask students to perform the corresponding inferential test. This process is then repeated over the course material without much deviation in the instructional method. In some cases, however, students can participate and benefit from well-designed classroom experiments [mcgowan2011]. This process can expose students to concepts such as randomization and let students see the specifics of experimental design through their participation. loy2021a has demonstrated that student participants often recalled their experiment in later concepts, showing some evidence that students can benefit from the hands-on experience.

In addition to using experiential learning broadly, a key aspect in the statistics classroom is teaching students to interpret data through graphs. Nearly 40 years ago, cleveland1984 began the process of establishing good practices for making graphs. While their findings have been replicated [heer2010], many areas in data visualization remain underdeveloped

that can benefit from the framework used by @cleveland1984. One such area is the projections of 3D data visualizations. The current mantra is to avoid 3D graphs when possible and studies around the 1990s seem to provide some valid skepticism of their use [@barfield1989; @fisher1997]. @barfield1989 showed that participants were sometimes more accurate using 3D graphs than 2D graphs, depending on the participant’s experience level. However, participants were generally more confident with their answers for 2D graphs. @fisher1997 observed a similar preference for 2D graphs over 3D graphs when extracting information, but found no preference for visual appeal for the type of graph. While these results provided valid skepticism toward the use of 3D graphs, the results are generalized to the projections of the 3D graphs. The area of “true” 3D graphs is largely unexplored, but advances in technology allow for easier access to explore the 3D projections of these graphs. @kraus2020 explored the 3D projections with the use of virtual reality and found that participants were more accurate at extracting values from 3D heatmaps at the expense of needing more time than 2D alternatives.

Because the use of “true” 3D graphs remains largely unexplored, it provides a unique opportunity to be used as an experiential learning opportunity for statistics students. Not only can students benefit from the exposure to different graph types, including “true” 3D graphs, but they can also experience a more authentic method of teaching, which is more likely to be beneficial to reinforce statistical ways of thinking.

In this paper, we used an experiential learning project that employed different graph types, including “true” 3D graphs, in an introductory statistics classroom environment and described its potential application as an educational tool. We presented students with a series of modules where they first participated in our experiment on different graph projections, followed by acquiring knowledge on the purpose of the experiment by reading an extended

abstract and watching a presentation of a pilot study of the same experiment.

12 Methods

12.1 Overview and Motivation

We introduced students enrolled in Introduction to Statistics (STAT 218) at the University of Nebraska-Lincoln to a graphics project that contained an experiment and progressively revealed components that illustrate the experiment’s research objectives. The project started by providing students with minimal information about the research objectives before revealing the scope of the experiment through an extended abstract and presentation. The goal of the graphics project was to observe how students think statistically about experiments as both participants and research consumers. Students were provided with mostly open-ended prompts throughout the graphics project, which allowed us to freely explore common themes in student responses without the limitation of preset choices.

12.2 Experiential Learning

The graphics project was split into two main components regarding the interaction of students with the material. In this section, we will discuss student interactions with the experiment. Students were first presented with a series of four modules presented from

the role of research participation. These modules contained the informed consent form, pre-experiment reflection, experiment participation, and post-experiment reflection. Within the informed consent module, students were informed that their data would be anonymized and that the experiment was exempt from the institutional review board (Project ID: 22579). While all students were given the option to participate in the graphics project, we were only able to collect responses when informed consent was obtained and if the student was 19 years of age or older. In the pre-experiment reflection, students were asked to write a paragraph about how they think the process of scientific investigation looks from the perspective of researchers and the general public. The experiment participation module asked students to paste the code generated upon completion of the graphics experiment, which is detailed in the next section. The generated code serves as a basic check to indicate whether or not students fully participated in the experiment. For the post-experiment reflection, students were asked five questions about the purpose of the experiment. These include questions on the hypotheses being tested, sources of error, variables of interest, and elements of experimental design.

After completing the experiment reflections, students moved to the reflection of the overall research objectives. Students were first directed to read an unpublished two-page extended abstract that we submitted as a contributed paper for the Symposium on Data Science & Statistics (SDSS). The extended abstract outlined the experiment’s purpose and procedures, but not the results from our initial pilot study. After reading the extended abstract, students were asked to write a paragraph about what they found clearer about the experiment’s purpose than when they were a participant. Finally, students were directed to watch a 12-minute pre-recorded presentation based on an abbreviated version given at SDSS [wiederich2023]. The video contained the same material as the extended abstract

and included the results from our pilot study. The presentation reflection asked students four questions about the experiment and how information was presented differently than in the extended abstract.

Except for the informed consent and experiment participation modules, all student responses were open-ended. Each question and its corresponding module can be found in Table 1.

Instructors for STAT 218 were recruited for Summer 2023 and Fall 2023 to administer the graphics project into their classroom. The instructors were given the option of administering the project as coursework material or extra credit, along with the liberty of grading at their discretion. Instructors for the June-July Summer 2023 courses did not have the abstract or presentation modules.

12.3 Data Analysis

Since nearly all of the student responses to the project modules are open-ended, the analysis of the project modules is qualitative. We will selectively extract student responses that demonstrate variability and repetitive themes in their understanding of the graphics experiment. For all modules, except the Post-Experiment reflection, students are asked to comment on the experiment without an objectively correct response.

For paragraph responses in the Pre-Experiment and Abstract reflections, bigram plots are used as a graphical analysis to display word pairs after removing stop words (e.g., “the” and “and”). These word pairs help illustrate common themes that students wrote about in their longer prompts. In the Post-Experiment reflection, the prompts have objectively correct answers but would require a subjective aggregation to indicate the level of correct-

ness. Instead, we opt to select at least one student response that we consider to be “most correct” and a couple of other responses that are either partially correct or entirely incorrect to illustrate the variability of the responses. The results of the graphics experiment are presented by @wiederich as an extension of a larger study comparing the dimensionality and projections of 2D and 3D bar charts.

12.4 Graphics Experiment

12.4.1 Constructing Stimuli

Based on Cleveland and McGill’s seminal work [–@cleveland1984] on graphical perception, participants were presented with a series of bar graphs where two bars are marked with either a circle or a triangle. The heights of the bars were chosen from the following equation:

$$s_i = 10 \cdot 10^{(i-1)/12}, \quad i = 1, 2, 3, \dots, 10 \quad (12.1)$$

where s_i represents a value given an integer i as defined above. The values s_i from Equation 12.1 were then paired such that the ratio of the smaller value to the larger value yield the ratios of 0.178, 0.261, 0.383, 0.464, 0.562, 0.681, and 0.825. Each bar graph has two groupings of five bars. Following the Type 1 and Type 3 graphs from the position-length experiment by @cleveland1984, the value pairs for each ratio were either placed in the first grouping on the second and third bars (adjacent) or placed on the second bars in each grouping (separated).

To explore the effect of dimensionality and projection of the bar graphs, we introduced the

following plot types: 2D digital, 3D digital (static), 3D digital (interactive), and 3D printed. There was no single software package that could create all four plot types, so we carefully constructed graphs from different software packages to be as similar as possible. The 2D digital plots were rendered with the `ggplot2` package [@ggplot2]. Microsoft Excel® was used to render the 3D digital (static) plots. The 3D digital (interactive) and 3D printed plots were created with OpenSCAD® [@kintelOpenSCADDocumentation2023], where the generated stereolithography (STL) files for the 3D digital (interactive) plots were rendered with the `rgl` package [@rgl].

12.4.2 Experimental Design

With 56 treatment combinations, we opted to use an incomplete block design to provide participants with 15-20 graphs. Kits of graphs were constructed so that five of the seven ratios are equally represented, resulting in 21 different kits. Within each kit, all graph types appeared for each ratio and the comparison type was randomly assigned. A visual layout of the experiment is shown in Figure 1. All kits received a unique identifier and a set of instructions for accessing the experiment.

A Shiny application [@shiny] was designed to administer the experiment. Students were directed to randomly select a kit of graphs and visit the Shiny application’s website linked on the instructions. For students enrolled in the online sections of STAT 218, the website was provided by the instructor and they were prompted in the application to select that they were an online participant; selecting online participation resulted in the 3D-printed plots being removed from the set of graphs presented to the participant. After students provided a kit identifier (if applicable), students were presented with graphs in a randomized

order. If the student marked that they were an online participant, the 3D-printed graphs were removed from their experiment lineup. Each graph asked the students to first identify the larger marked bar and then to guess the height of the smaller marked bar if the larger marked bar was 100 units tall using a slider widget. After completing the experiment, a completion code was generated for students to paste into the experiment participation module.

13 Results

13.1 Recruitment of Students and Instructors

We recruited 3 instructors for the summer and fall semesters in 2023. Each instructor offered the project as extra credit in their course and student participation was entirely voluntary. A total of 82 students participated in the project, and 9 students did not complete the entire project (Table 2).

13.2 Selected Responses from Experiment Participation

13.2.1 Pre-Experiment Reflection

In the first stage of the project, before participating in the experiment, students had limited information about the research objectives and we recorded how students thought about scientific research. Students were initially prompted with one question, “What do you think [the] process [of scientific investigation] looks like, from the perspective of a researcher, compared to what it looks like from the perspective of someone in the general public who is a consumer of scientific results?” The responses to this question allowed us to gather a

broad understanding of what students initially thought about the research process. Students generally understood the purpose of scientific research by connecting the ideas of hypothesis testing and publishing results. Students wrote about scientific research starting from the place of a question, followed by conducting an experiment and relaying the results to the public. For example, the most common phrase groupings include variations of “scientific research”, “data collection”, and “peer review”. A bigram plot of student responses to the Pre-Experiment reflection prompt is shown in Figure 2.

13.2.2 Post-Experiment Reflection

After participating in the experiment, students were provided prompts asking about the goals of the research objectives, which were to evaluate the errors of ratio judgments between 2D and 3D graph types under different comparison conditions in a randomized block design. Some students correctly identified parts of the questions asked in the post-experiment reflection, but most students typically missed some or most of what would be considered a correct answer.

We first asked students “What do you think the purpose of the experiment was?” For this question, a complete response would have included the comparisons of ratio judgments of the projections for 2D and 3D graphs. A student example of the correct response was “I think this experiment aimed to test if it was easier to compare two graphs in 2D or 3D.” Another student missed the comparison of 2D to 3D graphs, but was mostly correct with their response “I think the purpose of this experiment was for the researcher to gather data on how people perceive, interpret, and understand 3D graphs.” One student incorrectly thought that we were testing differences in demographics by responding “They could be

trying to determine how different genders, ages, etc. perceive the sizes of the bars in the graph. Demographics could make a pretty significant difference.”

We then asked students “What hypotheses might the experimenters have been testing?” A correct response would include testing the differences in errors of ratio judgments between graph types. One example of a student providing a correct response was given by stating “They might have been testing if a 2D model is easier to estimate its relative size to another when compared to a 3D model of it.” Other students replied with statements that would not be able to be measured from the experiment, with one student responding “How taking Statistics 218 effects how you can compare two groups” and another student saying “That 2d is preferred over 3d. It cleans up the data presentation.”

An important topic covered in STAT 218 is randomization, which we asked students with the prompt “What elements of experimental design, such as randomization or the use of a control group, do you think were present in the experiment? Why?” In the experiment, randomization was performed with the ordering and assignment of graphs into kits. One student perfectly described randomization by responding with “Randomization: The survey used an experimental design where in the survey there were different sets of 3D charts and maybe by a randomization process each participant was shown a different set of charts to see the differences in interpretations of the charts based on which set was assigned. Control Group: Since this survey aimed to only understand how participants interpret 3D charts without comparison to other chart types, then no control group was needed.” Another student was partially correct with a response of “Randomization was used because the ever person got a different graph.” Other students missed the use of randomization, with one student responding “random students in the stats class” and another saying “Randomization was not used because it was offered as an extra credit assignment in class.”

13.3 Selected Responses from Research Reflections

In this set of reflections, students first read the extended abstract, followed by watching the 12-minute presentation video. The abstract unveiled the scope of the study to students, many of whom did not realize the underlying complexities. Nearly all students responded with statements about gaining clarity about the purpose of the experiment and its role in testing the differences between 2D and 3D graphs. Students commented on how they now understood the purpose of comparing 2D and 3D graphs, and also the potential benefits that may stem from research, such as graphical accessibility to the visually impaired. A bigram plot of the student responses to the abstract reflection prompt is shown in Figure 3.

Lastly, more than half of the students (78.5%) responded that they preferred the presentation over the extended abstract when asked “If you had to hear about this study using only the extended abstract or only the presentation, which one would you prefer? Which one would be better for determining whether the experiment was well designed?” One student responded “I am a visual learner so I would have rather heard about in through the presentation. It also broke down the steps which is easier for me to understand. I think the presentation as a whole would be better for determining how the experiment is designed.” Another student said “Personally I like the abstract better. If I get confused on something it is so much easier to go back and reread to understand what is going on. If I ask myself questions about it, it is much easier to go back and find answers to the questions as well.”

14 Discussion

Taken together, our results support the idea that we achieved our goal of providing students of an introductory statistics course with the opportunity to reflect on active research. Students generally appreciated the progressively revealing nature of the graphics project, which is evident from the abstract and presentation reflections. When provided with the post-experiment reflections, students often either missed the research objectives of the experiment or had partially correct responses. However, this was expected since the design of the experiment exceeds the scope of experimental design taught in typical introductory statistics courses.

The abstract reflection received many responses indicating that students had moments of realization about the true nature of our research goals, which was further expanded with the presentation reflection prompts. Across all reflections, students were thoughtful, and sometimes amusing, with their responses and they were on the path of statistical thinking.

Having students participate in experiments is not uncommon, possibly attributed to the readily available convenient samples within academia and potential applications to introduce new course material [margaret1994]. However, these studies typically end for students after completing the experiment [mcgowan2011; zacks1998; fisher1997]. In our study,

we integrated the graphics experiment as a course project with the addition of reviewing research material.

What we found was that the graphics project found success within the recommendations of the GAISE College Report by using real data to illustrate the contextual purpose of scientific research [Carver]. Students were able to demonstrate their ability to think statistically through the series of reflections. The graphics project made use of real data, along with data collection, within the scope of an approachable and field-related topic, which allowed students to see how scientific research is conducted in the field of statistics.

A limitation of our study is the use of open-ended responses that do not objectively assess student learning. While the student responses were useful in gathering insight, the responses are widely varied and do not have direct comparisons of statistical thinking throughout the modules. Another limiting factor is the tiered layering of convenience sampling, with instructors being recruited before recruiting students, which impacts the generalization of our findings. Nonetheless, the 82 students who participated provided meaningful answers that displayed some level of statistical thinking throughout the graphics project.

Future studies could use a similar framework to conduct experiments on more typical 3-dimensional data, such as heatmaps. The use of graphical experiments in the classroom not only provides a readily available convenience sample but also adheres to the recommendations of the GAISE College Report [Carver]. With the framework we provided in this paper, we aim to make adjustments to further improve the graphics experiment and corresponding project as an experiential learning opportunity.

15 References

16 Figure legends

Figure 1: Visual display of the experimental design for students who participated in the 3D bar charts experiment. Kits of graphs were created by first choosing five ratios from nine available options (1). Each ratio then uses all graph types, with the exception of the 3D-printed graphs for online students (2). Finally, all graphs were randomly assigned to have the marked bars as adjacent or separated (3).

Figure 2: Bigram of student responses (n=82) to the pre-experiment prompt. Each line represents pairs of words that appeared together where each pair occurred at least twice. Students generally understood that science is about investigating research questions and collecting data.

Figure 3: Bigram of student responses (n=63) to the abstract reflection prompt. Each line represents pairs of words that appeared together where each pair occurred at least twice. Students generally focused on the research objective of comparing 2D and 3D graphs.

17 Tables

Table 17.1: Questions provided to students in each project module.

Reflection	Question	Prompt
Pre-Experiment	Q3	In this class, you'll be learning about the process of scientific investigation. What do you think that process looks like, from the perspective of a researcher, compared to what it looks like from the perspective of someone in the general public who is a consumer of scientific results? Write a paragraph (at least 3-5 sentences) about how you think science happens.
	Q5	What do you think the purpose of the experiment was?
Post-Experiment	Q6	What hypotheses might the experimenter have been testing?
	Q7	What sources of error are involved in this experiment?
	Q8	What variables were examined? For each variable, identify whether it was quantitative or categorical.
	Q9	What elements of experimental design, such as randomization or the use of a control group, do you think were present in the experiment? Why?

Abstract	Q10	What components of the experiment are clearer now than they were as a participant? What questions do you still have for the experimenter? Write 3-5 sentences reflecting on the abstract.
	Q11	How did the information you gained from the components of this project (participation, post-study reflection, extended abstract, presentation) differ?
Presentation	Q12	What components were emphasized in the presentation that weren't emphasized in the abstract? Why do you think that is?
	Q13	What critiques do you have of this study and its design? What would have made the study better?
	Q14	If you had to hear about this study using only the extended abstract or only the presentation, which one would you prefer? Which one would be better for determining whether the experiment was well designed?

Table 17.2: Number of valid student participants by semester.

Semester	Number of Sections	Number of Students
Summer 2023 (May-June)	1	17
Summer 2023 (July-Aug)	1	23
Fall 2023 (May-June)	1	42

Students under 19 years of age or did not consent were excluded from data collection. To comply with IRB, no demographic information was collected to keep students anonymous with their reflections.

18 Heat3d: A Study on 3D Heatmaps

STAT 992 Final Paper

19 Introduction

In the 20th century, advancements in technology made data visualizations increasingly more affordable and accessible to a broader population. The primary change in formal data visualizations was from hand-drawn charts to computer-rendered charts [tukey1965], yet other technological advances have allowed for data visualizations to enter other mediums. These include the ability to effectively use the 3-dimensional (3D) world around us with the novel use of 3D-printed charts. This newer type of visualization has gained a small amount of traction in recent years as a method of producing tangible charts.

At this time, there are few, if any, studies that evaluate 3D-printed charts for the purpose of displaying statistical information. This may in part be due to the cost of materials and rendering times as compared to charts produced on a computer. A single chart can take up to a day to print, limiting the ability to quickly produce visualizations. Given the nature of 3-dimensional data, viewing a 3D realization of statistical information in a 3D environment is a realistic scenario that could increase the ability for information extraction.

3D-printed visualizations have shown mixed or promising results across many disciplines. katsioloudis2018 showed no evidence of a statistical difference in the method of 3D renderings when tasked with drawing a cross-sectional of a dodecahedron. This demonstrated

that spatial awareness between computer-rendered and 3D-printed shapes is not largely different among engineering students. The use of 3D-printed maps for navigation for visually impaired persons showed positive feedback by @holloway2019, increasing the accessibility of navigation. In the clinical setting, 3D-printed anatomy structures were well accepted along with VR-glasses and 3D computer renderings [@muff2022]. With the rise of 3D-printed visualizations in scientific fields, we turn our attention to their use for statistical graphics.

19.1 Literature Review

An identical dataset across multiple chart types does not ensure that the data is perceived in the same way [@cleveland1984; @hofmann2012; @vanderplas2020]. One of the main factors to this phenomena is with how data is encoded into the chart. These encodings include, but are not limited to, placement along axes, lengths, areas, volumes, and color scales. For example, bar charts and pie charts are two common visualizations that have long since been the topic of debate [@eells1926; @croxton1927; @cleveland1984]. In the case of bar charts and pie charts, encodings are represented by lengths and angles, respectively. Inherently, the comparison of different chart types is a comparison of encodings due to the changes in how data is being displayed.

@cleveland1984 noted that estimates involving numerical accuracy may decrease when increasing dimensionality of the encoding, although this was not formally tested in their experiments. The reasoning is possibly due to Stevens' power law, a mathematical formulation of how magnitudes are perceived given different stimuli sources [@stevens1986]. The general form of the law is $\psi(I) = kI^\alpha$, where I is the magnitude of a stimulus, $\psi(I)$ is the perceived magnitude, k is a proportionality constant from the unit of the stimulus, and

α is the exponent from the type of stimuli used. Studies have estimated that lengths are perceived without bias (i.e., $\alpha = 1$), but areas and volumes tend to have skewed perceptions [cleveland1984]. This indicates that lower-dimensional charts might perform better when readers are tasked with extracting numerical estimates from the chart.

There are mixed results in regard to the use of 3D charts, mostly attributing to the purpose of third dimension. When the extra dimension does not convey meaningful information, estimates of accuracy decrease and solution times increase as compared to equivalent 2D charts [fisher1997; zacks1998; fischer2000]. The same increase in solution time is seen when the third dimension is utilized for displaying data, but can sometimes produce better error rates than 2D charts [barfield1989; kraus2020]. Additionally, when given the option of 2D or 3D charts for extracting numerical information, the 2D charts showed increased preference and confidence than their 3D counterparts [barfield1989; fisher1997]. It is worth noting that all of the studies listed use renderings of 3D charts and not physical 3D charts.

Formal studies involving true 3D charts are limited, and it is unclear if they follow the framework of existing theories on data visualizations. Unlike paper and computer rendered charts, constructing true 3D charts inherently contains many additional factors that could affect perception, such as chart materials, natural lighting, interactivity, and viewing distance. Some of these factors have already been shown to have an effect on computer renderings [tarr2001; wang2022].

We hypothesize that 3D charts in 3D environments will produce better information extraction than their computer rendered counterparts. Specifically, we will compare 3D-printed charts to digital 2D and 3D renderings. We suspect that this difference will hold across

multiple data sets and different magnitudes of stimuli. In this paper, we evaluate the accuracy of numerical estimations on true 3D charts by conducting a factorial experiment that assessed chart types and ratios of pairs of stimuli. We discuss the construction of the stimuli and how we closely matched the charts to compare 2D, 3D-digital, and 3D-printed renderings of heatmap data.

20 Methods

Our study is designed to evaluate and expand the literature on numerical estimation of 3D charts. In our study, our focus is on 2D and 3D heatmaps, which we carefully construct to ensure that differences are contributed to the dimensionality of the chart. All of our data and methods are publically available for open science and reproducibility at <https://github.com/TWiedRW/ch3-heat3d>. In this section, we discuss process of designing our experiment and participant recruitment.

20.1 Stimuli

We denote “stimuli” to represent the magnitude of our chosen values. In a 3D Cartesian space, the X- and Y- axes represent the coordinates of the stimuli, and the Z-axis represents the value for the stimuli. Each X and Y coordinate is represented by a square tile with a 1:1 aspect ratio. For a 2D space, the Z-axis is replaced by a color gradient scale. All stimuli and remaining randomly generated values range between 0 and 100 units. In this experiment, $X = 1, 2, \dots, 10$ and $Y = 1, 2, \dots, 10$.

The design of our experiment made use of the method of constant stimuli, where comparisons between stimuli are with respect to a stimuli that remains the same magnitude

[@kingdom2016 chap. 3]. We set the constant stimuli at 50 units. For stimuli between 50 and 100, we set the maximum stimuli value at 90 and equally partitioned the ratios of stimuli with the constant stimuli, $\frac{50}{90} = 0.556$ to $\frac{50}{50} = 1.0$, resulting in 4 varying stimuli values where 50 is the smallest value. The same ratios obtained with stimuli between 50 and 90 were used to create 4 stimuli varying between 0 and 50, where 50 is the largest value. Additionally, we also included a stimuli pair where both values are 50, resulting in nine total pairs of stimuli. All stimuli values can be found in Figure 20.1.

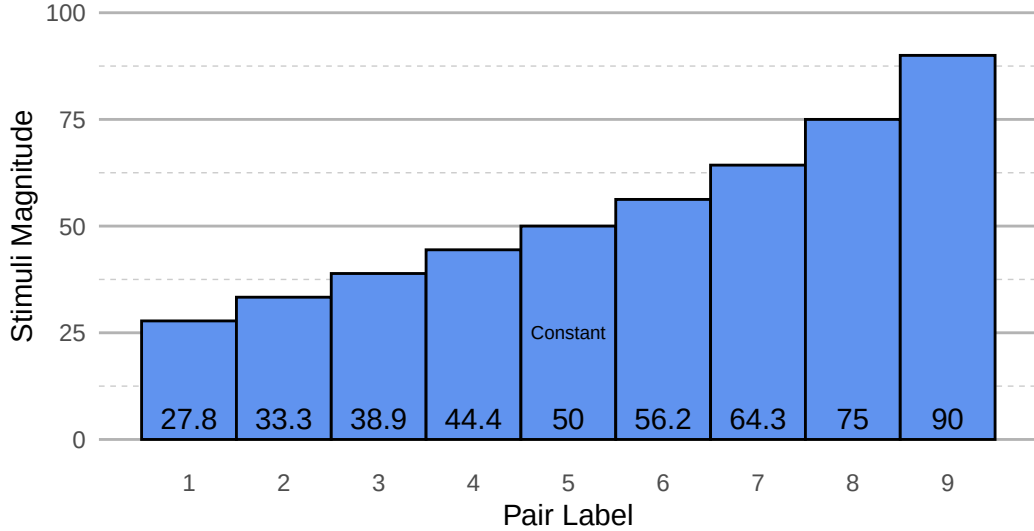


Figure 20.1: Values for stimuli in the heatmap experiment. All values are paired with the constant stimuli of 50 units, creating nine pairs of stimuli.

To generate non-stimuli random values, we used a mixture distribution of random uniform noise and a mathematical function to populate our coordinate grid. The mathematical functions are scaled between 0 and 100, $g(Z) = 100 \cdot \frac{Z - \min(Z)}{\max(Z) - \min(Z)}$. Two datasets were created for the experiment. The first dataset, called set 1, used the formula for the top half of sphere that is centered within our X and Y coordinate grid, $f_1(X, Y) = \sqrt{7^2 - (X - \bar{X})^2 - (Y - \bar{Y})^2}$, where $\bar{X}$ and $\bar{Y}$ are the averages of the X and Y coordinate ranges. The second dataset is calculated similarly using the formula for the bottom half of sphere, $f_2(X, Y) = \sqrt{7^2 - (X - \bar{X})^2 + (Y - \bar{Y})^2}$. Denoting Z as the random values,

$U(0, 100)$ as a random variable drawn from a continuous uniform distribution, and X, Y as heatmap coordinates, our random heatmap data is calculated in Equation 20.1 with $c = 0.3$. An example of varying c values is presented in Figure 20.2.

$$Z = c \cdot U(0, 100) + (1 - c) \cdot g(f_i(X, Y)) \quad (20.1)$$

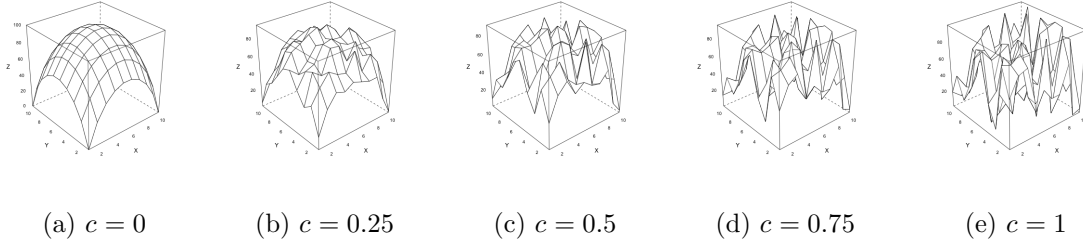


Figure 20.2: Mixture distribution of Equation 20.1 using the formula for the top half of a sphere. As c approaches 1, the distribution resembles uniform random noise.

The placement of stimuli values onto the randomly generated heatmap data was done via simulation to try to make their placement look as natural as possible. Twenty random heatmaps were generated. For each heatmap, the non-constant stimuli was placed onto the coordinate such that the difference between the stimuli and randomly generated value is minimized. The constant stimuli was then placed onto a coordinate with a Manhattan distance of three or four that minimizes the difference between the constant stimuli and randomly generated values, where the Manhattan distance is given by $|X_i - X_j| + |Y_i - Y_j|$ for stimuli i and j . To ensure that stimuli placement is evenly position across the heatmap, the count of stimuli was computed separately across the X and Y axes. For example, in Data Set 1, the X -axis has four stimuli in $X = 1$, three stimuli in $X = 2$, and so forth. Chi-squared statistics were calculated for each axis and the heatmap with the smallest average Chi-squared statistic was selected as the final dataset. A visual inspection of this process

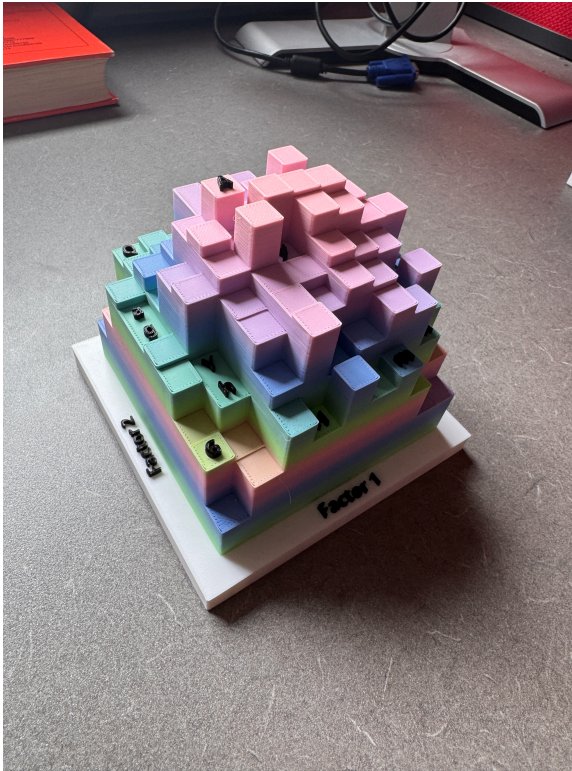
showed that the stimuli were not clustered in any one area of the chart and that the stimuli look natural with respect to the random mixture distribution.

20.2 Charts

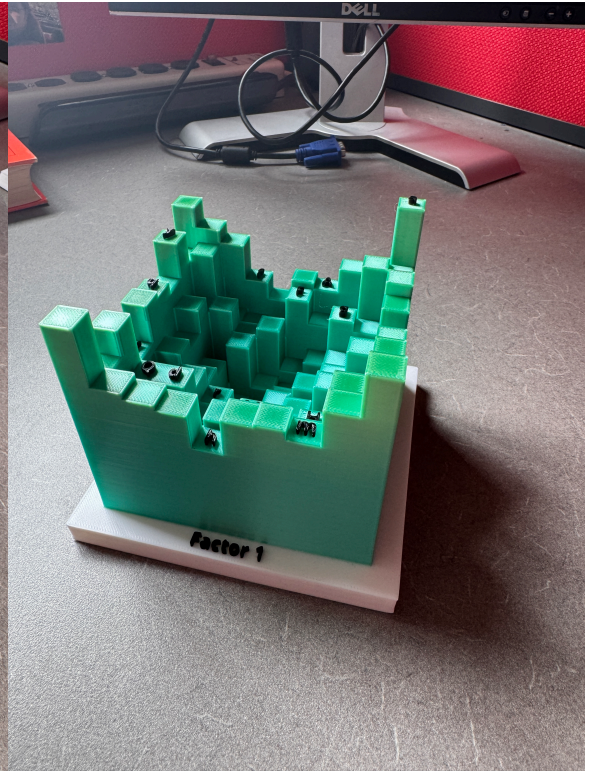
Three types of charts were considered for this study: 2D-digital (2dd), 3D-digital (3dd), and 3D-printed (3dp). We constructed these charts so that they are as similar as possible, but inherent difference between dimensionality led to artistic decisions that attempt to focus solely on the dimensionality of the charts. The process of creating the charts is discussed in this section.

The 3D-printed charts were rendered with OpenSCAD [kintelOpenSCADDocumentation2023]. To include plot text, a 120mm by 120mm by 10mm base was created with a solid color that was either white or black. Cells of the heatmap measured 10mm by 10mm, resulting in a heatmap that is 100mm by 100mm and is centered on the base. The upper bound of the height of heatmap values is 100mm, where 1-unit in the heatmap data is represented by 1mm of height on the heatmap. Once rendered, the heatmap was saved to 3D Manufacturing Format (3mf) and Standard Triangle Language (stl) files. A variety of solid and gradient filaments were used to print the output files from OpenSCAD. An example of the 3D-printed chart is shown in Figure 20.3.

To closely match the 3D-digital chart to the 3D-printed chart, multiple stl files were created for each colored component and combined with the RGL package [rgl]. The base was rendered with white smoke (#F5F5F5) to slightly contrast with the default white background color (#FFFFFF). Heatmap tiles were rendered with cyan (#74CCFF) and text labels were rendered with black (#000000). Lighting was fixed at 45 degree angles at two



(a) Solid filament of Data Set 1



(b) Solid filament of Data Set 2

Figure 20.3: 3D printed heatmaps.

opposite corners of the chart. The end result was a near perfect replica of the 3D-printed charts, with the exception of different heatmap tile colors, where an example is given in Figure 20.4.

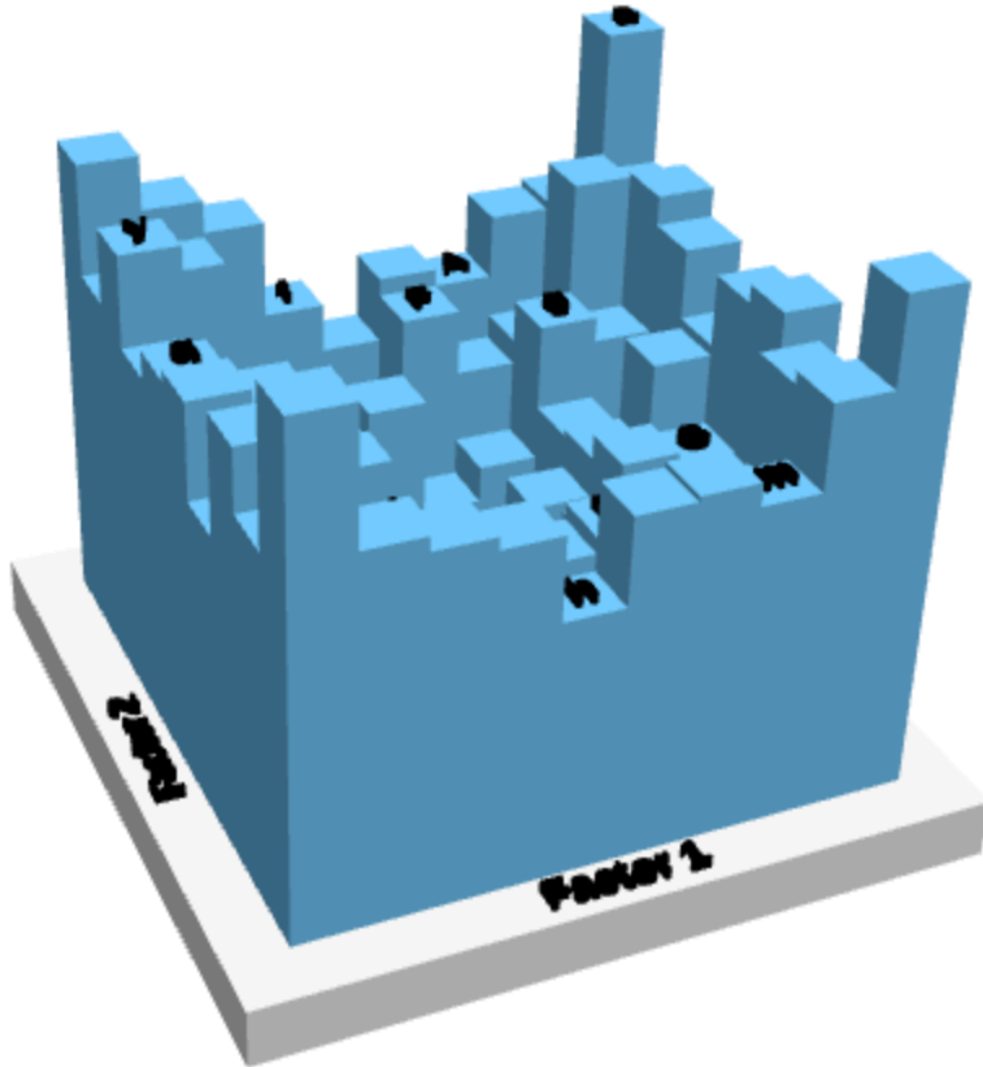


Figure 20.4: RGL rendering of Data Set 2 for 3dd charts.

Unlike the 3D charts, the 2D charts needed a different encoding to convey heatmap values. The 2D heatmaps were created with `ggplot2` [ggplot2] using `geom_tile()`. Fill colors for the cells use a color gradient from Blue Zodiac (#0C2841) to Malibu (#66D9FF), which

were selected from a color picker using shadows on our initial 3D-printed charts. The color interpolation was performed with the `scale_fill_gradient()` function from the `ggplot2` package.



Figure 20.5: Color palette for 2D-digital charts. The colors are interpolated from #0C2841 to #66D9FF, which are the the colors of the lighting conditions for a 3D-printed chart created with cyan filament.

20.3 Subject Recruitment

Participants were recruited from all STAT 218: *Introduction to Statistics* sections at University of Nebraska-Lincoln from June 2025 to Spring 2026. The experiment was incorporated into the curriculum as a project that allowed students to get hands-on experience with statistics. Throughout the project, students were also given a series of reflections to evaluate their perspectives on statistics. The first reflection had gathered students' thoughts on the scientific process before participating in the experiment. After the experiment participation, students reflected on what they thought the experiment was measuring. Lastly, students were given a written article and video presentation of the experiment and asked how their view of the experiment changed from their initial perspectives of the experiment's purpose.

To assess whether students completed the experiment or not, the experiment produced a completion code for students to submit to Canvas. The completion code was created by concatenating three words with dashes (e.g., palm-raising-creatively). Words for the completion code came from the movie transcripts of the Star Wars prequel movies (Episodes

1, 2, and 3) as presented by movies.fandom.com. The movie transcripts were tokenism and filtered to remove colorful language¹, stop words², and fictional words³. Instructors were provided an R script to check if their students completed the experiment, but the completion codes were not saved with any identifying information that could link experiment responses to student reflection responses.

20.4 Experimental Design

Our experiment was designed with a 3 x 2 x 9 treatment structure. Media type is our main interest, with 2D-digital, 3D-digital, and 3D-printed charts. To ensure that results are not confounded with datasets, we used two datasets to create the heatmaps. A total of 9 stimuli pairs were placed into each dataset. The order of treatments was given so that media and dataset combinations were grouped together randomly in the sequence and stimuli pairs were randomized within the groupings.

Due to practical constraints, stimuli pairs were incompletely blocked. A full factorial design would result in 54 trials per participant, which could lead to a decrease in quality responses [herzog1981]. Therefore, we selected four out of the nine possible stimuli pairs to create incomplete blocks. This resulted in 18 blocks for a balanced incomplete block design. Within each block, media type is fully crossed with dataset. Using the incomplete block structure, participants completed a total of 24 trials, which is more practical than the full factorial design. Blocks were chosen for each participant by using probabilities that are proportional to the inverse counts of used blocks, meaning

¹<https://www.cs.cmu.edu/~biglou/resources/bad-words.txt>

²tm package [tm]

³@web2_dict

We measured three responses – two questions for each trial and one question for each media by dataset combination. For each trial, participants are initially asked which stimuli in a pair is larger or if the stimuli are the same value. Next, they were asked to estimate the magnitude of the smaller stimuli if the larger stimuli represents 100 units, which is a subtractive process [a@veit]. After each grouping of media and dataset, participants were presented with a modal dialog window asking them to rate their confidence on a 5-point Likert scale for their answers of the previous group of charts.

20.4.1 Shiny Application

A Shiny application [a@shiny] was developed to administer the experiment. The application consisted of five sections: informed consent, demographics, practice, experiment, and wrap-up. The entire application is designed to be completed in approximately 30 minutes.

The Shiny application started with the informed consent screen, allowing participants to select if they are a STAT 218 Student and if they agree to the data collection. Participants had to select a data collection option to continue. After submitting their data collection response, a completion code was generated and saved on the last page of the application. A copy of the informed consent is available in our GitHub repository.

If participants agreed to have their data collected, they were presented with the demographics section. This section asked participants to use drop down menus to specify their age, gender identity, highest education level, and a question about how their participation in the experiment is graded. The last question was a text box and asked participants to specify their favorite movie and/or actor. Demographic information was combined with the application start time and completion code to create a hash for an anonymized participant

identifier using the `rlang` package [RStudio].

Once a participant completed the demographics page or selected “No” to the data collection question, participants were given a practice page. A modal dialog window was initially shown with instructions, and users could display this window again at any point during the practice. The practice consisted of four questions: two trials from 2dd charts and two trials from 3dd charts. Each practice trial showed the correct solution after the participant submits their trial. After all practice trials were completed, another modal dialog window was displayed to ask participants if they have access to the 3D-printed charts. This was necessary since the experiment was given to potentially online sections of STAT 218 who did not have access to the physical charts.

The experiment page was presented to participants after completion of the practice trials Figure 20.6. Each page contained two questions – one question for identifying which stimuli is larger and another question for estimating the value of the smaller stimuli if the larger stimuli is 100 units. If participants selected that the stimuli are the same value, then the slider was automatically placed at 100 units. Each trial has the slider initially placed at 50 units. An option was provided for displaying the coordinates of the stimuli pair on a 2D heatmap, but was left unchecked by default. For 3dd charts, the number of interactive clicks was also recorded. A trial could only be submitted if the first question is answered and if the slider was moved at least once.

After completing all trials, participants were given the completion code and informed to copy the code to Canvas since they will not have access to the code after closing out of the experiment.

Trial 1 of 24

Group 1 of 6

Which of the following values is larger?

☐ F
☐ f
☐ The two values are the same.

If the larger value you selected above represents 100 units, how many units does the smaller value represent?

0

50

100

Submit

☐ Select this checkbox if you need help identifying the two values on the chart.

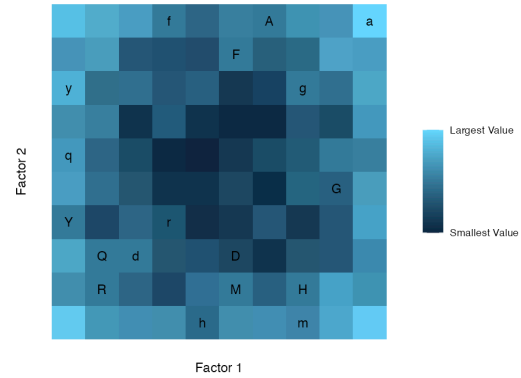


Figure 20.6: Graphics experiment page. Each trial contained two questions about a pair of stimuli, and a check box that displays the location of the stimuli if needed.

20.4.2 Method of Analysis

The design of our experiment is a balanced incomplete split-plot with replication, where media type and dataset is the whole plot and stimuli pair is the split plot. A similar design was presented by @mandal2020 for a single replicate. For our experiment, the ANOVA table is in Table 20.1.

Table 20.1: ANOVA table for analysis of a single replicate of the balanced incomplete split-plot.

Source	Degrees of Freedom
Block (B)	18-1
Media type (M)	3-1
Data Set (D)	2-1
Whole-plot error = $B \times M \times D$	$(18-1)(3-1)(2-1)$
Stimuli Pair (P)	9-1

Source	Degrees of Freedom
$P \times M$	$(9-1)(3-1)$
$P \times D$	$(9-1)(2-1)$
$P \times M \times D$	$(9-1)(3-1)(2-1)$
Split-plot error = $B \times M \times D \times P$	$(18-1)(3-1)(2-1)(9-1)$

The first question for each trial was to indicate which value within a stimuli pair was larger, or if they were the same value. Since a participant could either answer this question correctly or incorrectly, a Binomial generalized linear mixed model [atstroup] was fit to the responses.

To measure numerical accuracy of participant responses for the second question in each trial, we used the absolute value of the difference between the participant estimate of the smaller stimuli value and its true value. By using the absolute value of the error, we measure how far off a participant's estimate was rather than the direction of error. For numerical accuracy, we fit a linear mixed model. If the responses do not fit well to a Gaussian distribution, other distributions, such as the log-normal or gamma distributions, will be used for a generalized linear mixed model.

$$Y = |\text{Guess} - \text{Actual}|$$

In addition to linear mixed models, we used the bootstrap distribution of mean errors as described by @cleveland1984 to provide empirical claims for the effect of dimensionality. A common issue in experiments involving perception is that random noise often covers the

effects of interest, so the empirical claims may provide additional insight into the effect of chart dimensionality (XXX).

After each media type and data set grouping, participants were given a question to rate their confidence on their answers. Here, we use descriptive statistics and visualizations to assess confidence across our treatment factors.

21 Limitations

Our study was designed to evaluate the accuracy of numerical estimation of stimuli pairs across three types of charts. However, numerical accuracy is not the only deciding factor in what makes a good chart design. Other factors include comprehension, recall, and overall preference [fisher1997; vanderplas2020]. Given the lack of current research into 3D-printed statistical graphics, we hope to provide some baseline insights into their use as a statistical tool.

We did not account for potential visual illusions in the construction of the heatmap data. A phenomenon seen in statistical graphics studies is that surrounding values may attribute to participant estimates [zacks1998; vanderplas2015]. In our study, the placement of stimuli was so that the stimuli looked naturally positioned within the structure of the data, resembling patterns typically seen in spatial data. However, consistency of neighboring values was not accounted for in this experiment and would be worth having a follow up study to measure the effect of surrounding values on 3D-printed charts.

Lastly, we wanted to construct heatmaps so that they closely resembled each other across dimensionality. The 3dd charts are identical to the 3dp charts, with the exception of heatmap colors. Here, the issue is that 3D heatmaps are not typically constructed by

compiling stl files, but rather using packages such like `lattice` package [@lattice] to display the data. Due to parameter settings and code limitations, it may not be possible to get typical 3D renderings of charts to appear visually identical to 3D-printed counterparts.

22 Conclusion

Our study provides insight into the novel use of 3D-printed statistical graphics. We hope that these charts may be useful for educational purposes, accessibility of visually impaired persons, or recreational use of statistical communication.